Modeling Routing Behavior for Vacant Taxi Cabs in Urban Traffic Networks

Xianbiao Hu[1], Song Gao[2], Yi-Chang Chiu[3], Dung-Ying Lin[4]

[1]Doctoral Student, Department of Civil Engineering and Engineering Mechanics, University of Arizona, Tucson, USA, xbhu@email.arizona.edu
[2]Assistant Professor, Department of Civil and Environmental Engineering, University of Massachusetts at Amherst, Amherst, USA, songgao@ecs.umass.edu
[3]Associate Professor, Department of Civil Engineering and Engineering Mechanics, University of Arizona, Tucson, USA, chiu@email.arizona.edu
[4]Assistant Professor, Department of Transportation & Communication Management Science, National Cheng-Kung University, Tainan, Taiwan, dylin@mail.ncku.edu.tw

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ABSTRACT

Taxi cabs account for a significant portion of traffic in many Asian cities, and yet, the route choice of taxi drivers remains an active research area in the literature. The routing objectives of a taxi driver vary depending on the state of taxi occupancy. If a taxi is occupied by customers, then a least-cost path is usually sought. Several existing paradigms in the literature are related directly to such a routing objective. What is not well understood is the taxi driver's route choice behavior when a taxi is vacant, which is likely to be driven by a rather different objective. In this research, we propose a vacant taxi routing model in which taxi drivers are assumed to minimize expected search time for customers when making routing decisions at intersections. A probabilistic dynamic programming formulation of the problem and the solution algorithm are presented. We conduct numerical analysis on a hypothetical network resembling the traffic network structure in the City of Taipei. The outcomes of this case study show that the proposed model exhibits realistic and reasonable properties. This research has shed light on routing decisions of taxi drivers, which will directly support area-wide traffic management.

Keywords: vacant taxi cab, routing model, probabilistic, dynamic programming, traffic management, traffic modelling

1 INTRODUCTION

Taxi traffic generally accounts for a significant portion of urban traffic in many Asian cities. A survey by the Ministry of Transportation and Communications (MOTC) in Taiwan reported that there are more than 88,000 taxis in Taiwan *(1)*, of which 73.5% regularly cruise on streets for passengers[1]. These taxi cabs on average travel 136.1 km daily, of which 54.4 km (equivalent to 40% of the total distance) are vacant distance, i.e., with no passengers. This translates into a waste of approximately 90 million liter of gas annually. The average 3.6 vacant hours per day also directly impact the urban traffic congestion level. In the urban area of Hong Kong, taxis currently accounts for about 25% of the overall traffic stream, and in some critical locations, the percentage increases to 50–60% *(2)*. Another study also found that for all the taxis in Shenzhen, China, 41.6% of the total distance and 49.2% of the total time are in the vacant state *(3)*. These statistics underscore the substantial amount of vehicle hours and distance traveled by vacant taxis, and highlight the need to understand better how these vacant taxis navigate in the urban traffic network in order to contribute to effective urban traffic management strategies.

There are generally two types of taxi service market: the dispatch market and the cab stand and street hail market *(4)*. In a dispatch market, the dispatcher (or fleet operator) dictates which taxi should go and pick up a given customer, and sometimes even which route to take when the dispatching is based on a GIS. Therefore a taxi driver's personal routing objective is usually not put into effect in a dispatch market. In a cab stand and street hail market, the taxi drivers are autonomous decision makers and decide where to go and which routes to take.

The routing objectives of a taxi driver vary depending on the state of taxi occupancy. If a taxi is occupied by customers, then a least-cost path is usually the objective sought. Several existing paradigms in the literature are directly related to such a routing objective *(2, 5-7)* . What is not well understood is a taxi driver's route choice behavior when a taxi is vacant, in which case the driver's behavior is likely to be driven by a rather different objective from minimal travel time. It is intuitive to assume that an occupied taxi driver seeks to arrive at the destination in the least amount of time or distance as expected/required by the customer[2], similar to the objective of a commuter in his/her own car. If a taxi is vacant, the objective generally is to minimize the elapsed time until picking up the next customer (search time) and the decision to be made is which downstream link to run at each intersection. If a taxi starts from a starting link where the driver is aware of the low probability of getting customers (e.g. the taxi drops off a customer at a location where returning, or ongoing, customers are few at the time), the driver will intend to find his way to where he knows the customer arrival rate may be high. He/She

[1] 26.5% of taxies wait in line at transit terminals, airports, or provide charter services.

[2] In this study, we don't consider customers who hail a taxi for city sightseeing purpose nor drivers who dishonestly seek to increase a fare by traveling the long way around, particularly for strangers in that city.

knows the general direction to go but is still making local route choice decisions so that he/she remains on the lookout for customers while moving toward that high probability area.

It is intuitive to postulate that (1) a taxi driver aims to maximize his/her chance of reaching a taxi-seeking customer when making a turning decision at an intersection; (2) the outcomes of past trials form the "experience" or "belief" about customer availabilities for specific geographic areas at given time periods of the day; (3) such an experience or belief affects the routing decision; and (4) the experience or belief will be updated regularly by outcomes of new trials. In this research, we propose a probabilistic dynamic programming vacant taxi routing model that is conceived based on the above behavior propositions.

We conduct both analytical derivation and numerical analysis on a hypothetical network inspired by the traffic network structure in the City of Taipei. The contribution of this study is that it shows that our proposed methods offer a promising modelling framework for capturing vacant taxi driver's routing behavior and the study sheds light on modelling routing decisions of taxi drivers, which is critical for developing strategies such as path-based traffic simulation or origin-destination estimation in a region with a high taxi traffic percentage.

The rest of this paper is organized as follows: Section 2 reviews taxi related literature, including the studies that are critical to this research. Section 3 discusses the modelling framework and approach in depth, and Section 4 presents the case study results. Section 5 concludes this research.

2 LITERATURE IN TAXI RESEARCH

A rather wide and diversified body of research about taxis exists in the literature, studying various aspects of the taxi industry and operating environment. These taxi-related research efforts could be generally classified in four categories: (1) how to improve the overall system performance from the fleet operator's standpoint or how to integrate novel technologies to improve customer service, (2) passenger's decisions and preference, (3) taxi driver decisions and behavior, and (4) policy issues from system perspective considering demand-supply relationships.

The first category of research includes examples such as Lee and Cheng *(8)* investigating a new taxi business model that utilized smart phones to improve operational effectiveness. The ridesharing concept was explored in several studies. Tao and Wu *(9)* explored the internal and external factors affecting a customer's acceptance of taxi-sharing services. Tao *(10)* also presented algorithms for dynamic ridesharing matching process for taxi commuters in the Taipei area of Taiwan. The focus of this research was primarily on matching passengers and assigning taxis so that the distance/time traveled of both passengers and taxis can be minimized. A related study on taxipooling was also discussed *(11, 12)*. Further, Chang et al. *(13)* proposed a data mining process to predict taxi demand based on time, weather, and taxi location. An automatic multi-agent dispatch system was proposed by Kiam et al *(14)*. Traffic characteristics - including operating condition and vacancy before and after the Beijing Olympic Games were reported *(15)*.

Examples of literature in the second category include a case study in Taiwan that modeled consumer preference for a global positioning system (GPS) based taxi dispatching service by Hwang et al. *(16)*. Sheu et al. *(17)* studied taxi hailing in vehicular ad-hoc networks. Tao and Wu *(9)* aimed to understand passengers behavior response to the dynamic ridesharing services, and field trial of dynamic taxi-sharing services was conducted in their research.

Taxi regulation and policy issues were extensively discussed in several critical studies in which the general problem was modeled and analyzed in an equilibrium analysis framework. A network model was built to describe the demand and supply equilibrium of taxi services under certain fare structures and fleet size regulations, and was able to determine a number of system performance measures at equilibrium and predict the effects of alternative regulations on system performance *(5)*. Yang and Wong *(6)* developed a model to describe how vacant and occupied taxis will move in a road network to search for customers and provide transportation services. They concluded that the average taxi utilization decreases sharply with the number of taxis operating, and the higher the taxi utilization, the larger average customer waiting time. Wong et al. *(7)* discussed the different classes of customers, multiple taxi modes and customer hierarchical model choice. They proposed the idea that there are several classes of customers with different values of time and money, and several modes of taxi services with distinct combinations of service area restrictions and fare levels. A simultaneous mathematical formulation of two equilibrium sub-problems for the model was built in the paper.

While considerable research has studied taxis from the perspective of the fleet operator, the passenger and the system policy-maker, research considering a taxi driver's travel decision is relatively sparse. A survey-based behavioral science study focused on general driving behavior (e.g. excessive speeds and carelessly changing lanes) as related to risk taking personalities (needs for more tension, risk and adventure in their lives) observed from driver record data. *(18)*. Wong developed a mathematical model that is based on the absorbing Markov Chain approach to describe taxi movement, taking into account the stochastic searching processes of taxis in a network. He used a logit choice model to specify the local searching behavior, and estimated the origin-destination (O-D) demand of taxi passengers using a logit model with a choice of taxi meeting point. A set of non-linear equations were used to solve the problem *(19)*. Yang and Leung *(20)* proposed an equilibrium model to characterize the bilateral searching and meeting between customers and taxis on road networks, and proposed a meeting function to characterize the search and meeting frictions that arise endogenously as a result of the spatial feature of the area and the taxi-customer moving decisions.

General anecdotal observations from the authors' interviews with taxi drivers in Taipei conducted in 2011 can be summarized as follows. (1) Taxi drivers adjust their business hours according to experienced customer arrival pattern, which means that supply is adaptive to the temporal pattern of demand. (2) Experienced drivers know the spatial pattern of customer arrivals and that residential and commercial areas are major demand generators, respectively, in AM and PM peak hours. (3) Once a taxi driver drops off a customer, he/she may choose to circulate around, but if the timing and/or area is not right and they do not find customer in 15-20

min, they will return to areas where they know more customers may show up. This means they somehow have mental recognition of a demand generation pattern spatially and temporally. (4) Taxi drivers are conscious about competition. (5) There are heterogeneous customer types and taxi drivers have preference in favor of different types of customers *(21)*. For example, some areas will generate lots of customers near mid-night (e.g. night clubs), and some taxi drivers will go for those customers, but others will avoid such a customer because there is a higher chance of running into troublesome customers.

In summary, the characterization and modeling of the routing decision by vacant taxi drivers remain to be further researched despite of a solid body of literature in the general topic.

3 RESEARCH APPROACH TO THE VACANT TAXI ROUTING PROBLEM

The taxi driver travels in a traffic network $G = (N, A)$ where N is the set of nodes $\{1, 2, \dots, n\}$ and A the set of link. Each link (i, j) is uniquely defined by the source node i and sink node j, and has travel time t_{ij} . Divide link (i, j) into multiple segments, each with the same travel time δ. Customers arrive at any segment of link (i, j) following a Poisson process (negative exponential inter-arrival time) with a rate of $\lambda_{ij}\delta$, where λ_{ij} is the base arrival rate over a unit of travel time on link(i, j). The base rate is determined by the commercial and residential land use patterns around the road link in question. We do not explicitly model multiple taxis on the same link; rather, we scale down the customer arrival rate to reflect multiple taxis' competition for customers. The link arrival rate λ_{ij} is for one taxi. The customer arrival processes at different links are assumed independent in one link from another.

We do not model the within-link details and assume that a taxi will pick up a customer on a segment of a link (i, j) if the arrival of the next customer at this segment happens during the taxi's trip on this segment. This is effectively assuming that all customers arriving at the segment will form a single queue at the end of the segment and wait for the taxi. Since in reality customers can arrive at any location along the segment, the value δ should be chosen such that this assumption is valid. On the one hand, δ should not be too large. First, if multiple taxis are competing for customers, the taxi will have to be fairly close to the arrived customer to pick her up. Secondly, if the taxi has passed an arrived customer, the distance between them must be small enough for the taxi to slow down and the customer to catch up. On the other hand, δ cannot be too small. Consider the extreme case where δ goes to zero, such that the taxi and the customer have to arrive at a location along the link at exactly the same time for a pick-up. The chance of such event approaches zero, given the continuous nature of the arrival along the time and space dimensions. Clearly this is not a realistic result. Without any field data, we postulate that δ can take a value between 5 to 10 seconds in a busy Asian city based on the authors' personal experience.

The elapsed time between a taxi leaving the downstream node of an origin link and picking up a customer is defined as the search time. It is a random variable, as the customer

arrivals are random. We assume the taxi driver aims to minimize the expected search time from the downstream node of any given origin link $(k,i) \in A$.

In the real world, U-turns are at times found to be forbidden for many links, both on freeway and at intersections, and left-turns at intersections will take more time than through and right turn movements. For this study, a penalty function $PN(k,i,j), \forall (k,i) \in A,\ (i,j) \in A$ is introduced to penalize U-turn and left-turn movements.

3.1 Search Time on a Single Link

The modeling starts from formulating the calculation of the expected search time on a single link, g, and then extend the results to a general network. For the sake of simplicity, we omit subscripts and denote the travel time on this link as t and the base arrival rate λ. We derive the expected search time on a link by backwards recursion starting from the last segment, assuming the number of segments is an integer, which is not a restrictive assumption if δ is small enough. Given that the taxi has reached the beginning of the last segment, there are two possible outcomes by traversing the segment: picking up a customer or not. The search time given no pick-up is simply the segment travel time δ. Denote the elapsed time between the taxi leaving the beginning of the last segment and the next customer arrival at the segment as x_{-1}. The expected search time on the last segment given a pick-up, $E[x_{-1}|x_{-1} \le \delta]$, is derived next.

$$
\begin{aligned}
E[x_{-1}|x_{-1} \le \delta] &= \frac{\int_0^\delta x_{-1}\lambda\delta e^{-\lambda\delta x_{-1}} dx_{-1}}{P(x_{-1} \le \delta)} = \frac{\int_0^\delta -x_{-1} de^{-\lambda\delta x_{-1}}}{P(x_{-1} \le \delta)} \\
&= \frac{-x_{-1}e^{-\lambda\delta x_{-1}}|_0^\delta + \int_0^\delta e^{-\lambda\delta x_{-1}} dx_{-1}}{P(x_{-1} \le \delta)} = \frac{(-x_{-1}e^{-\lambda\delta x_{-1}} - \frac{1}{\lambda\delta}e^{-\lambda\delta x_{-1}})|_0^\delta}{P(x_{-1} \le \delta)} \\
&= \frac{-\delta e^{-\lambda\delta^2} - \frac{1}{\lambda\delta}e^{-\lambda\delta^2} + \frac{1}{\lambda\delta}}{P(x_{-1} \le \delta)}
\end{aligned} \tag{1}
$$

The expected search time on the last segment, g_{-1}, is then

$$
\begin{aligned}
g_{-1} &= E[x_{-1}|x_{-1} \le \delta]P(x_{-1} \le \delta) + \delta P(x_{-1} > \delta) \\
&= -\delta e^{-\lambda\delta^2} - \frac{1}{\lambda\delta}e^{-\lambda\delta^2} + \frac{1}{\lambda\delta} + \delta e^{-\lambda\delta^2} = \frac{1}{\lambda\delta}(1 - e^{-\lambda\delta^2})
\end{aligned} \tag{2}
$$

Let x_{-2} denote the elapsed time between the taxi leaving the beginning of the second to the last segment and the next customer arrival at the segment. The expected search time on the last two segments (not the second to the last segment), $g_{-2,-1}$, can then be derived as follows:

$$g_{-2,-1} = E[x_{-2}|x_{-2} \le \delta]P(x_{-2} \le \delta) + (\delta + g_{-1})P(x_{-2} > \delta)$$
$$= -\delta e^{-\lambda\delta^2} - \frac{1}{\lambda\delta}e^{-\lambda\delta^2} + \frac{1}{\lambda\delta} + \left(\delta + \frac{1}{\lambda\delta}\left(1 - e^{-\lambda\delta^2}\right)\right)e^{-\lambda\delta^2} \quad (3)$$
$$= \frac{1}{\lambda\delta}(1 - e^{-2\lambda\delta^2})$$

The recursion is continued until the expected search time on the last $\frac{t}{\delta}$ segments is found, which is the expected search time on the whole link:

$$g = \frac{1}{\lambda\delta}(1 - e^{-\lambda\delta t}). \quad (4)$$

Finally the probability that the taxi does not pick up a customer on the link can be derived as:

$$p = P\left(x_{-t/\delta} > \delta\right) \dots P(x_{-2} > \delta)P(x_{-1} > \delta) = e^{-\lambda\delta^2} \dots e^{-\lambda\delta^2}e^{-\lambda\delta^2} = e^{-\lambda\delta^2 t/\delta} = e^{-\lambda\delta t}. \quad (5)$$

3.2 Optimality Condition

The problem is a special case of the stochastic shortest path problem studied in *(22)*, where it is formulated as an infinite horizon dynamic programming problem with a cost-free termination state, which effectively is a finite-horizon problem with a random number of stages (traversed links). In the following we show the specialization with respect of the taxi case.

We present a link-based formulation to allow for turn penalties. A link $(k, i) \in A$ defines a system state, whose control set is the set of all outgoing links from node i, $A(i)$, indicating the taxi driver's decision to take a given next link.

For a given control $(i, j) \in A(i)$, let x_{ij}be the elapsed time between leaving node i and the next customer arrival on link (i, j). According to Eq. (4), the expected search time on link (i, j) is

$$g(i, j) = \frac{1}{\lambda_{ij}\delta}(1 - e^{-\lambda_{ij}\delta t_{ij}}). \quad (6)$$

The state transit probability from link (k, i) to (i, j) given control (i, j) is the probability that no pick-up happens on link (i, j), and according to Eq. (5) ,

$$p_{ij} = e^{-\lambda_{ij}\delta t_{ij}}. \quad (7)$$

We create a virtual termination state *d* for the whole network that is connected to any link. Trivially, the state transit probability from link (k, i) to the termination state given control (i, j) is 1 - p_{ij}.

Let $S^*(k, i)$ be the minimum expected search time for a taxi starting at the end of link (k, i). Assume that regardless of the chosen routing policy and the origin link, there is a positive probability that the taxi will pick up a customer after a finite number of link traversals. A necessary condition for this assumption to hold is that there is no cycle with zero arrival rate. According to Proposition 7.2.1 of *(22)*, $S^*(k, i)$ are the solution of Bellman's equations $\forall(k, i) \in A$. Note that the cost-to-go term associated with the termination state, $(1 - p_{ij})S^*(d)$ is omitted as $S^*(d) = 0$.

$$S^*(k,i) = \min_{(i,j)\in A(i)} \left[g(i,j) + p_{ij} S^*(i,j) + PN(k,i,j) \right], (k,i) \in A \quad (8)$$

A stationary policy μ^* is optimal if and only if for every state (k,i), $\mu^*(k,i)$ attains the minimal in Bellman's equation (8), namely

$$\mu^*(k,i) = arg \min_{(i,j)\in A(i)} \left[g(i,j) + p_{ij} S^*(i,j) + PN(k,i,j) \right], (k,i) \in A \quad (9)$$

3.3 Solution Algorithm

Equations (8) and (9) can be solved by policy iteration. The algorithm works as follows:

1. Start with a feasible stationary policy $\mu^z, z = 0$.
2. Evaluate the expected search time of μ^z for each origin link *(k,i)* by solving the system of simultaneous equations:

$$S^z(k,i) = g\left(i, \mu^z(k,i)\right) + p_{i,\mu^z(k,i)} S^z\left(i, \mu^z(k,i)\right) + PN(k,i,\mu^z(k,i)), (k,i) \in A, \quad (10)$$

   with $|A|$ unknowns $S^z(k,i), (k,i) \in A$. This can be solved exactly using Gaussian elimination with a computational complexity of $O(|A|^2)$.
3. Perform a policy improvement, which computes a new policy μ^{z+1} as

$$\mu^{z+1}(k,i) = arg \min_{(i,j)\in A(i)} \left[g(i,j) + p_{ij} S^z(i,j) + PN(k,i,j) \right], (k,i) \in A \quad (11)$$

4. $z = z + 1$ and repeat the above two steps until $S^{z+1}(k,i) = S^z(k,i), \forall (k,i) \in A$.

An illustrative example is presented in the next section to demonstrate the proposed methodology.

4 CASE STUDY

A case study was conducted to illustrate the solution properties of the proposed model. A real world network was constructed but certain assumptions about model inputs were made so that the research efforts were more concentrated on the analysis and results interpretations than collecting actual data. Although this may affect the real-life applicability of our proposed approach, these assumptions could be relaxed with modest changes. Further discussions in this regards are offered in the final section of this paper.

We chose one of the busiest districts in the Taipei metropolitan area (depicted in Figure 1) to be our case study region. Within the area, there are many central administrative institutions (i.e. Taipei City Government [1] and Taipei City Council [2] et al.) with each generating a nexus for business traffic in this network. In addition, National Dr. Sun Yat-Sen Memorial Hall [3], a common and popular tourist attraction, is located near the center of this network. The landmark skyscraper Taipei 101 [4], one of the tallest buildings in the world, is also located in this area. Together with a line-up of flagship retailers and boutique stores around and the Hotel of Grand Hyatt Taipei [5], this area has always drawn a vast amount of tourist traffic. Moreover, passengers from the five major mass rapid transit (MRT) stations and the Song Sang Railway Station [6] may need to transfer from one mode to another at nearby facilities to reach their final destinations.

It is understandable that business traffic, tourist traffic and transfer services attract a large amount of taxi drivers circulating for customers. In light of several considerations at this stage of research, the customer arrival rates used in this case study are assumed values. The base customer arrival rates for most link segments are assumed to be random values between 0 and 0.5 per minute except the rates for areas near Taipei 101, which are assumed to be higher at 0.8 per minute per link segment, and 0.6 for areas near the Taipei subway lines. These two areas with higher arrival rates are highlighted in Figure 1. The rectangle stands for Taipei 101 and nearby area, and the oval stands for arterials along the Taipei subway line). In reality, the arrival rate for each link may be highly correlated with the land use and business activities associated with the link. Such correlation can be further derived from land use/commercial activities data in future research.



Figure 1: Taipei Case Study Network

Besides the arrival rates, another major input to the minimum expected search time problem is the travel time on each link. We produced such data from a dynamic traffic assignment (DTA) model, DynusT *(23)*, yet any other traffic model capable of assigning demand and producing link travel times can also be used. The resulting network consists of 755 nodes, 2332 links and 11 traffic zones. Due to the time-independence nature of this problem, all time-dependent link travel times produced by DynusT need to be aggregated into one single average value. After applying the aforementioned algorithm for iteration of the route selection policy, the links chosen in the optimal policy from every origin link are shown as the thicker lines in Figure **2**. Every upstream link has exactly one leaving downstream link as its optimal next link.

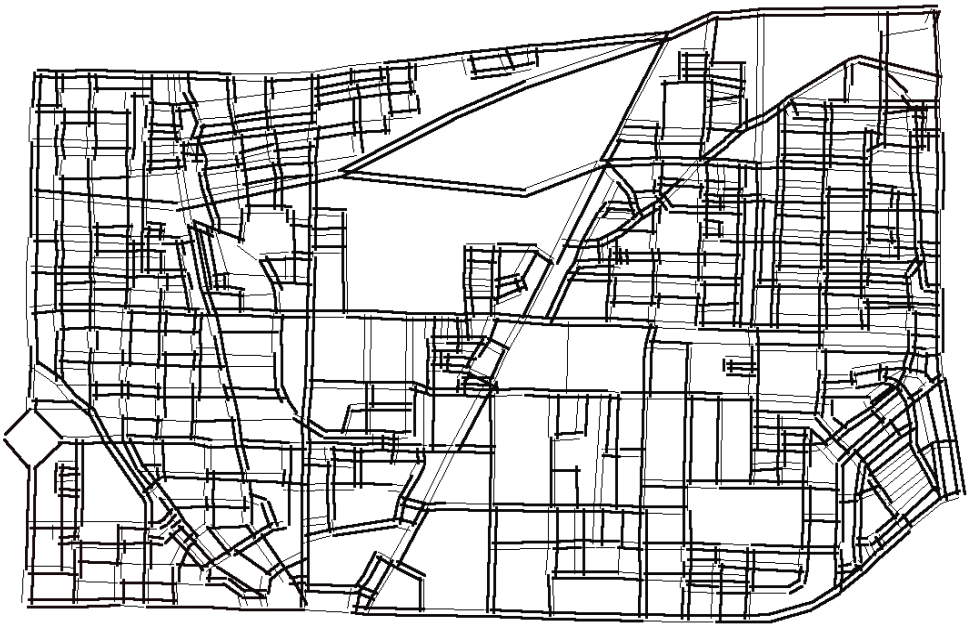


Figure 2: Vacant Taxi Routing Model Solution

With further investigation into the solution for each origin link one can generally identify two types of solutions – (1) a solution consisting of a closed loop or (2) a solution consisting of a cruising path and a subsequent closed loop. In the former type, a taxi starts from a link, goes through a set of links and comes back to the same link. A closed loop means that the taxi driver will keep driving along the cycle and looking for customers until he/she picks up a customer.

Examples of the closed loops are shown in Figure 3. These circled areas correspond to the busy areas in reality in Taipei City. For example, area 1 is the MRT Taipei City Hall station, area 2 is Taipei 101 section 5, XinYi Rd, area 6 is the MRT Zhonxiao Dunhua station, and area 9 is the Taipei Municipal Library Yung-Chung Branch. It is intuitive that taxi drivers prefer to stay in those busy areas and keep looking for customers until he/she picks up one.

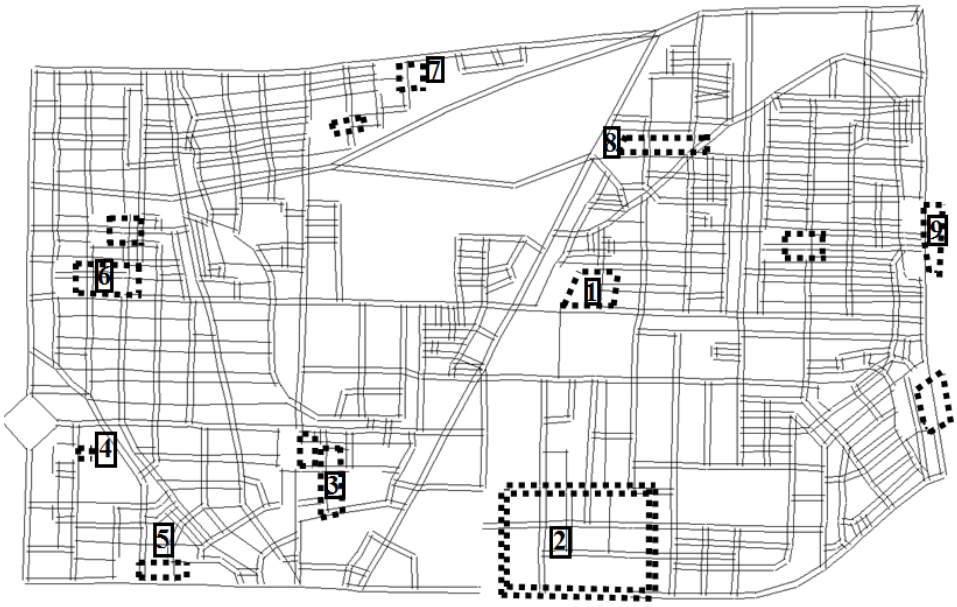


Figure 3: Optimal Policy Consisting of a Closed Loop

The second type of solution consists of an initial cruising path and a subsequent closed loop. In this case, the taxi driver may start from an area with lower customer arrival rate, then traverse a number of links, and finally enter a cycle in the areas with a higher customer arrival rate until finding a customer. In all, both solution forms exhibit a common property in that regardless of the starting link of a taxi, the driver will always aims to arrive at the area with higher demand and loop for customers in the vicinity until picking up a customer. Examples around the Taipei 101 area and the Taipei subway line area are shown in Figure 4.



Figure 4: Optimal Policy Consisting of Cruising and entering a Closed Loop

To further validate the above interpretation, we show the base arrival rates along an optimal policy from one given origin link in the Taipei 101 area in Table 1. The taxi starts from link 771 ->616, which is somewhat far away from that area, and then it makes a right turn and drives all the way to the Taipei 101 area. The links set after node 634 forms a closed loop. From Table 1, one can clearly see that the base arrival rates for all links prior to arriving at node 634 are considerably lower than those following node 634 (0.8). All the links following node 634 form a closed loop in this particular solution as shown in Figure 4.

Table 1: Link Base Arrival Rate for an Example of Optimal Policy

Node	Link	Base Arrival Rate
771	77100616	0.13
616	61600622	0.25
622	62200617	0.44
617	61700624	0.49
624	62400619	0.38
619	61900626	0.26
626	62600627	0.46
627	62700634	0.49
634	63400635	0.8
635	63500632	0.8
632	63200633	0.8
633	63300499	0.8
499	49900687	0.8
687	68700500	0.8
500	50000697	0.8
697	69700027	0.8
27	2700696	0.8
696	69600695	0.8
695	69500026	0.8
26	2600634	0.8
634	63400635	0.8
635	Continue to circulate	

The Policy – Base Arrival Rate – Travel Time relationships are plotted in Figure 5, with X-axis being the base arrival rate, Y-axis being the travel time, and Z-axis the optimal route selection policy. The upper layer (Z = 2) includes all the links chosen as the policies, and the lower layer (Z=1) stands for non-chosen links.

Generally speaking, most arrival rates are in the range of 0 and 0.5 with a few at 0.6 and 0.8. Per the previous discussions in the experiment setup section, these high arrival rate links are near Taipei 101 and subway stations. Link travel times reported by DynusT range from 0.1 to

about 1.0 minutes. One can further observe from the optimal policy solution that all links except one with arrival rate greater than or equal to 0.6 are included in the optimal routing policy. This is reasonable because links with higher arrival rates ought to be included in the optimal policy. Moreover, for a link, the higher the base arrival rate or travel time is, the more likely the link is chosen as part of an optimal policy. This result matches with Equation (7), that the probability of no pick-up happening on link (i, j) is inversely correlated to the base arrival rate and travel time.

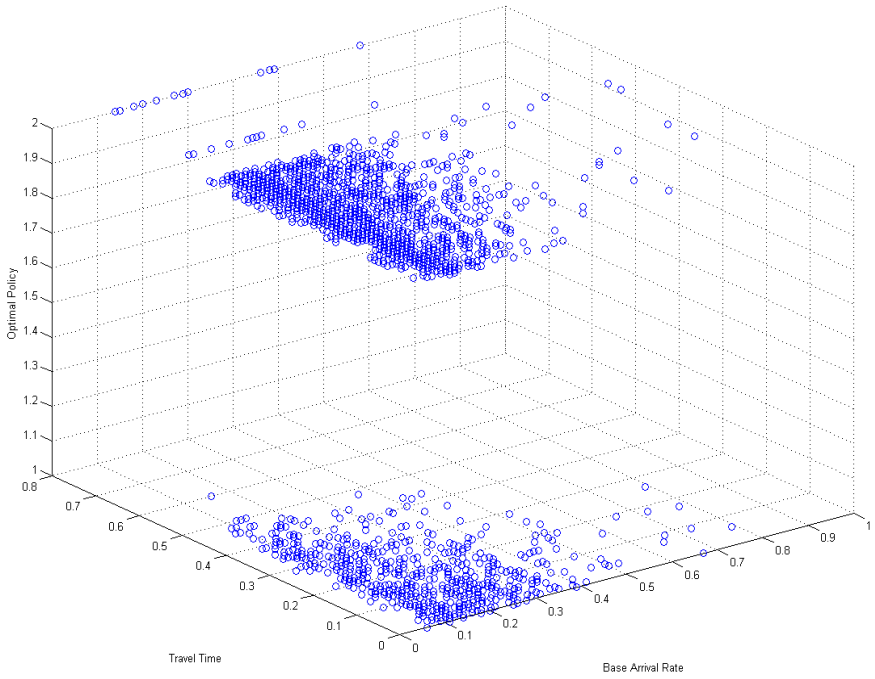


Figure 5: Policy – Base Arrival Rate – Travel Time Relationships

Figure 6 shows the relationship between base arrival rate and chosen policy. The 755 links in the Taipei network are divided into several groups according to their base arrival rate values. The Y-axis is the number of links chosen in the optimal policy divided by the total number of links in that group. The displayed trend indicates that a link with a higher base arrival rate may become more attractive to taxi drivers and is more likely to be chosen in the routing policy. According to the modeling results, in the selected area with higher base arrival rate, all links (100%) in the Taipei 101 area are chosen in the optimal policy and 25 out of 26 links (96%) in the Taipei Subway line area are chosen. It is noteworthy that this observation is a property of the model solution. Although this study uses only the assumed arrival rates the solution possesses rather intuitive and reasonable properties. The realism of the proposed model can be further validated with real data in the future.

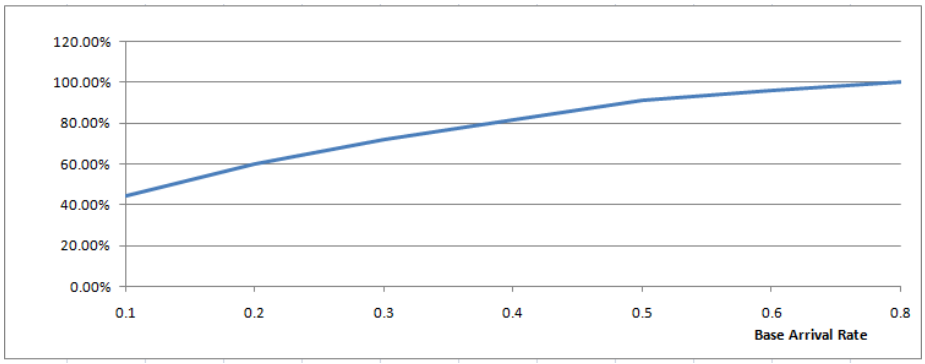


Figure 6: Relationship between Base Arrival Rate and Chosen Policy

5 CONCLUDING REMARKS

In this research, we propose a probabilistic dynamic programming vacant taxi routing model that is aimed at modelling a vacant taxi driver's routing behavior. We conduct both analytical and numerical analysis on a hypothetical network resembling the traffic network structure in the City of Taipei. The outcome of the numerical analysis indicates that two types of solutions are achieved by the model and both types include a loop. In one type, vehicles will start and end in the loop until the taxi finds the customer. In the other type, in which a taxi starts from a low customer arrival rate link, the optimal policy would be to move toward the high arrival rate area to start the circulation. Links with a higher arrival rate have a higher chance to be included in the solution. In all, the numerical analysis sheds light on routing decisions of taxi drivers, which will directly support area-wide traffic management. The future research focuses on two dimensions: (1) calibration of the proposed model vis-a-vis survey data from taxi drivers, and (2) estimation of the customer arrival rate data using land use, taxi fleet, or real-time traffic proxy data from the case study area. The emerging technology that helps taxi drivers reduce vacant circulation time will also be incorporated to improve the realism of the proposed model. A longer-term research direction is to formulate a dynamic model in which link travel times and arrival rates may be time-dependent.

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